



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Laban 84

Product name: 2,4-D-dimethyl ammonium 84% W/V SL

Product use: Herbicide

Relevant identified uses of the substance/mixture and uses advised against:

Commonly used as herbicide and secondarily a plant regulator.

2. HAZARD IDENTIFICATION

This product unlikely to cause harmful effects under normal conditions of handling and use. Can cause harmful effects if not handled and stored as per requirement.

Potential Human Health Effects:

Likely Routes of Exposure: Inhalation, eye and skin contact.

Inhalation,

Overexposure may cause upper respiratory tract irritation and symptoms similar to those from ingestion.

Eye contact,

Causes irreversible eye damage. Vapors and mist can cause irritation.

Skin contact,

This product is considered a minimal skin irritant and is not a dermal sensitizer. Overexposure by skin absorption may cause nausea, vomiting, abdominal pain, decreased blood pressure, muscle weakness, muscle spasms.

Ingestion,

Harmful if swallowed. May cause nausea, vomiting, abdominal pain, decreased blood pressure, muscle weakness, muscle spasms.



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Medical Conditions Aggravated by Exposure:

Inhalation of product may aggravate existing chronic respiratory problems such as asthma, emphysema or bronchitis. Skin contact may aggravate existing skin disease.

Potential Environmental Effects:

Toxic to aquatic invertebrates.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: 2,4-D-dimethylamine salt (DMA)

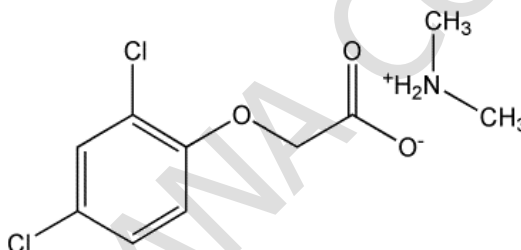
Common name: 2,4-D-dimethyl ammonium

Chemical class: Chlorophenoxy acid or ester

Empirical formula: $C_{10}H_{13}Cl_2NO_3$

Molecular mass: 266.13

Structural formula:



Ingredients and composition:

Component	CAS No.	% W/V
Active ingredient		
- 2,4-D-dimethyl ammonium	2008-39-1	84
Other ingredient	-	16

4. FIRST AID MEASURES

Description of first aid measures

In case of skin contact:

Remove contaminated clothing. Wash skin with water. If symptoms develop .obtain medical attention. Contaminated clothing should be laundered before re-issue.

In case of eyes contact:

Irrigate promptly with plenty of water. If problem persists obtain medical attention.

In case of Ingestion:

If swallowed induce vomiting by giving a tablespoon of salt in a glass of warm water. Repeat until vomit fluid is clear. If problem persists obtain medical attention.



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In case of Inhalation:

Remove patient from exposure. Obtain medical attention if ill effects occur.

Antidote or treatment:

Gastric lavage, Quinidine sulphate to counter myotonia and ventricular fibrillation, supportive and symptomatic treatment.

5. FIRE FIGHTING MEASURES

Flash point & Method:	Combustible liquid
Upper/lower flammability or explosive limits:	Not pertinent
Auto-ignition temperature:	Not pertinent
Explosive properties:	See 'Fire Fighting Measures'
Oxidizing properties:	Strong oxidizers

FIRE FIGHTING HAZARDS& PROCEDURES:

Extinguishing media

Water spray, foam, dry powder or carbon dioxide.

Fire protective Equipment

A self-contained breathing apparatus and suitable protective clothing should be worn if involved in fire with other materials.

Special Fire Fighting Procedures

Firefighters should wear self-contained breathing apparatus and full fire-fighting turn out gear. Dike area to prevent runoff and contamination of water sources. Dispose of fire control water later.

Unusual Fire and Explosion Hazards

If water is used to fight fire, contain runoff, using dikes to prevent contamination of water supplies. Dispose of fire control water later.

Hazardous Decomposition Materials (Under Fire Conditions)

Under fire conditions, may produce gases such as hydrogen chloride, nitrogen oxides, and carbon oxides.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal precautions, protective equipment and emergency procedures

Wear appropriate protective gear for the situation. See Personal Protection information in Section 8.



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Environmental precautions

Prevent material from entering public sewer systems or any waterways.

Do not flush to drain. Large spills to soil or similar surfaces may necessitate removal of topsoil.

The affected area should be removed and placed in an appropriate container for disposal.

Methods and material for containment and cleaning up

Methods for Containment:

Dike spill using absorbent or impervious materials such as earth, sand or clay. Collect and contain contaminated absorbent and dike material for disposal.

Methods for Clean-Up and Disposal:

Pump any free liquid into an appropriate closed container. Collect washings for disposal.

Decontaminate tools and equipment following cleanup. See Section 13: DISPOSAL CONSIDERATIONS for more information.

7. HANDLING AND STORAGE

Precautions for safe handling:

Do not get in eyes or on clothing. Avoid inhaling vapor or spray mist. Remove saturated clothing as soon as possible and shower. Wash hands before eating, drinking, chewing gum, using tobacco or using the toilet. Remove clothing immediately, if pesticide gets inside. Then wash thoroughly and put on clean clothing. Remove Personal Protection Equipment immediately after handling this product. Wash the outside of gloves before removing. As soon as possible, wash thoroughly and change into clean clothing.

Conditions for safe storage:

Always store pesticides in a secured warehouse or storage building. Store at temperatures above 32°F. If allowed to freeze, rewarm to 40°F, remix thoroughly before using. This does not alter this product.

Containers should be opened in well-ventilated areas. Keep container tightly sealed when not in use. Do not stack cardboard cases more than two pallets high. Do not store near open containers of fertilizer, seed or other pesticides. Do not contaminate water, food or feed by storage or disposal.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering Controls:

If this container is five gallons or more in capacity, do not open pour product from this container. A mechanical system (such as a probe and pump or spigot) must be used for transferring the contents



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of this container. If the contents of a non-refillable pesticide containers are emptied, the probe must be rinsed before removal.

Personal Protective Equipment

Wear protective equipment to comply with good occupational hygiene practice.

Eyes/Face Protection Requirements:

To avoid contact with eyes, wear face shield, goggles or safety glasses with front, brow and temple protection. An emergency eyewash should be readily accessible to the work area.

Skin Protection Requirements:

To avoid contact with skin, wear long pants, long-sleeved shirt, socks, shoes and chemical-resistant gloves. An emergency shower should be readily accessible to the work area.

Respiratory Protection Requirements:

Not normally required. If vapors or mists exceed acceptable levels, wear NIOSH approved air-purifying respirator with cartridges/canisters approved for use against pesticides.

General Hygiene Considerations:

Personal hygiene is an important work practice exposure control measure and the following general measures should be taken when working with or handling this material:

- 1) Do not store, use and/or consume foods, beverages, tobacco products, or cosmetics in areas where this material is stored.
- 2) Wash hands and face carefully before eating, drinking, using tobacco, applying cosmetics or using the toilet.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance:	Liquid
Color:	Brown
Odor:	Aliphatic amine
pH:	8 to 9
Melting point / freezing point:	Not applicable
Boiling point:	Not applicable
Vapor pressure:	0.4 mmHg at 320°C (- 1.1°C)
Density:	1.2 to 1.24 g/ml
Solubility in water:	Miscible
Partition coefficient:	log Kow = 2.58-2.53 (pH 1)



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10. STABILITY AND REACTIVITY

Hazardous reactivity:

Reactive to alkaline compounds

Conditions to Avoid:

This product should be kept in a cool place, preferably below 30°C.

Incompatibilities:

No particular incompatibilities.

Fire Decomposition:

Carbon dioxide, and if combustion is incomplete, carbon monoxide and smoke. Nitrogen and its compounds, and under some circumstances, oxides of nitrogen. Occasionally hydrogen cyanide gas. Hydrogen chloride gas, other compounds of chlorine. Water. Carbon monoxide poisoning produces headache, weakness, nausea, dizziness, confusion, dimness of vision, disturbance of judgment, and unconsciousness followed by coma and death. Hydrogen cyanide poisoning signs and symptoms are weakness, dizziness, headache, nausea, vomiting, coma, convulsions, and death. Death results from respiratory arrest. Hydrogen cyanide gas acts very rapidly; symptoms and death can both occur quickly.

Polymerization:

This product is unlikely to undergo polymerization processes

Hazardous decomposition products:

Toxic and irritating hydrogen chloride or toxic gases may form in case of fire.

11. TOXICOLOGICAL INFORMATION

Toxicity class WHO (a.i.): II

Information on Acute toxicity for end product, Tetraconazole:

LD50 (Oral, rat): 500 mg/kg

LD50 (Dermal, rat): > 4932 mg/kg

LC50 (Inhalation, rat, 4h): >1.36

Skin irritation (rabbit): Not irritant to skin in rabbits.

Eye irritation (rabbit, 48 h): Severely irritating to eye in rabbit

Skin sensitization: Non sensitizer in Guinea pig



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12. ECOLOGICAL INFORMATION

Persistence and Degradation: Half-life in soil is less than 7 days.

Toxicity

Data referred to the mixture, Tetraconazole:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h): > 100 mg/L

Daphnia,

LC50 (21 days): 235 mg/L

Aquatic plants – Acute toxicity:

Algae (*Selenastrum capricornutum*),

EC50 (5 days): 33.2 mg/L

Birds – Acute toxicity:

Japanese quail,

LD50: 668 mg/kg

Wild ducks,

LD50: > 1,000 mg/kg

Pigeons,

LD50: > 668 mg/kg

Pheasants,

LD50: 472 mg/kg

Mallard ducks,

LC50: > 5620 mg/L

Arthropod – Acute toxicity:

Honey bee,

LD50: 104.5 µg / bee

Effect on effluent treatment: No Data

13. DISPOSAL CONSIDERATIONS

Do not contaminate water, food or feed by storage or disposal.

Waste Disposal Methods:

Pesticide wastes are toxic. If container is damaged or if pesticide has leaked, contain all spillage. Absorb and clean up all spilled material with granules or sand. Place in a closed, labeled container for proper disposal. Improper disposal of excess pesticide, spray mixtures, or rinsate is a violation of Federal law and may contaminate ground water. If these wastes cannot be disposed of by use



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according to label instructions, contact your State Pesticide or Environmental Control Agency, or the Hazardous Waste Representative at the nearest EPA regional office for guidance.

Container Handling and Disposal:

Do not reuse empty container. Triple rinse (or equivalent), adding rinsate to spray tank. Then offer for recycling or reconditioning, or puncture and dispose of in a sanitary landfill, or by other procedures approved by State and local authorities. Plastic containers are also disposable by incineration, or if allowed by State and local authorities, by burning. If burned, stay out of smoke.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN number:	3082
UN shipping name:	Environmentally Hazardous Substance, Liquid, N.O.S.
Transport hazard class:	9
Packing group:	III
Label required:	"Keep Away From Food"
Marine pollutant:	Yes

15. REGULATORY INFORMATION

Symbol: Xn

Label elements

Risk phrases:

R 22 Harmful if swallowed.

R 20/22 Harmful by inhalation and if swallowed.

16. OTHER INFORMATION

This data sheet is prepared in accordance with directive 91/155/EEC

This product should be stored, handled and used in Accordance with good industrial hygiene practices and in conformity with any legal regulations.

The information contained herein is based on the present state of our knowledge and is intended to describe our product from the point of view of safety requirements. It should not; therefore, be construed as guaranteeing specific properties. Recipients of our product must take responsibility for observing existing laws and regulations.